MEMVLN: Streaming Episodic Memory for Vision-Language Navigation

Supplementary Material

A Additional Implementation Details

This section complements the implementation details in the main paper with the training protocol and memory configuration used for the default MEMVLN configuration.

Data Statistics. We summarize the training data statistics in Tab. A.1. Each instruction-conditioned demonstration is treated as one episode. For simulation training, the number of navigation samples counts the step-wise action-prediction targets after action merging; the trajectory summarization data is derived from the same episodes. For real-world post-training, we collect 24 demonstrations using an Intel RealSense D455 RGB-D camera mounted on the G1 robot for training. The D455 RGB sensor has a nominal horizontal field of view of 90°. Its RGB frames are captured at 1280×720 and resized to 640×480 before model processing to match the simulation input resolution.

Streaming Episode Sampler. Training a stateful memory policy requires temporally coherent data loading. Standard random sampling breaks an episode into unrelated steps, preventing the memory bank from being updated in the same chronological order encountered at inference time. We therefore employ a streaming episode sampler that assigns complete episodes to each device and feeds the steps within each episode sequentially. This protocol allows each device to maintain continuous memory streams while supporting distributed training and task mixing.

Default Memory Configuration. Following the memory module formulation in the main paper, the memory bank stores the visual feature f_t of each observation together with its timestep t and odometry z_t. We denote the maximum number of retained frame entries as N_b. Each frame is spatially compressed to at most N_f patch tokens using ToMe [1]. A new observation is first appended to the bank. If this increases the bank size to $N_b + 1$, frame merging consolidates the two consecutive entries with the highest average patch-level cosine similarity, reducing the bank size by one and restoring the capacity constraint. The memory perceiver then uses N_m learnable queries to retrieve a fixed-size set of memory tokens m_t. Since each query produces one output token, N_m is also the number of memory tokens injected into the vision-language model (VLM). The default configuration uses $N_f = 100$, $N_b = 100$, and $N_m = 300$. Additional settings are summarized in Tab. A.2.

Simulation Training. As described in the main paper, we adopt Qwen3-VL-4B [2] as the VLM backbone and StreamVGGT-1B [3] as the geometry encoder. The visual encoder and geometry encoder remain frozen. We finetune the large language model (LLM) with LoRA [4], while training the memory module and geometry adapter from scratch. More details are summarized in Tab. A.3.

Real-world Post-training. The real-world demonstrations record both the model prediction and the human-corrected executed action. We post-train the policy using the same token-level cross-entropy objective as in simulation, while assigning a larger sample weight to steps requiring correction:

$$\mathcal{L}_{\mathrm{PT}} = \frac{\sum_t w_t \mathcal{L}_t}{\sum_t w_t}, \qquad \mathcal{L}_t = -\log p(a_t^* \mid m_t, h_{(t-k):t}, \mathcal{T}). \tag{1}$$

We set w_t to 3.0 for corrected non-stop actions, 0.5 for actions accepted without correction, and 2.0 for stop actions. This correction-weighted behavior cloning objective focuses adaptation on states where the deployed policy requires human intervention.

Table A.1: **Training data statistics.** Navigation samples count step-wise action-prediction targets after action merging. Simulation sample counts are rounded to the nearest thousand.

Setting	Dataset	Training Data		RGB Input		
		Episodes	Nav. Samples	Resolution	HFOV	Camera
Simulation	R2R Train	10,819	~334K	640×480	90°	Habitat-Sim
	RxR Train	19,996	~1.02M	640×480	90°	Habitat-Sim
Real-world PT	G1 demonstrations	24	1,053	640×480	90°	Intel RealSense D455

Table A.2: **Default memory module configuration.**

Hyperparameter	**Setting**
Local observation window k	16
Max tokens per frame N_f	100
Memory bank capacity N_b	100
Memory token count N_m	300
Frame consolidation	Sim-based merging
Spatial embedding	Fourier features
Temporal embedding	Sinusoidal embedding
ToMe layers	2
ToMe attention heads	16
Perceiver layers	2
Perceiver attention heads	16

Table A.3: **MEMVLN training hyperparameters.**

Category	**Setting**
Hardware	$8 \times$ A100 80G GPUs
Precision	BF16
Training epochs	1
Per-device batch size	1
Gradient accumulation steps	1
Effective batch size	$1 \times 8 \times 1 = 8$
Optimizer	AdamW
Base learning rate	2×10^{-5}
Weight decay	0.05
Warmup steps	400
LoRA rank / alpha	128 / 256

Table B.4: **Ablation studies on memory token count N_m and memory bank capacity N_b.** Default configuration uses $N_m = 300$ and $N_b = 100$. Higher is better for OSR, SR, SPL, and nDTW; lower is better for NE.

Ablation	N_b	N_m	R2R Val-Unseen				RxR Val-Unseen			
			NE↓	OSR↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	nDTW↑
Number of memory tokens										
Default	100	300	**5.19**	**61.2**	**55.0**	**50.5**	**5.59**	**55.4**	**48.4**	**64.6**
More memory tokens	100	600	6.75	53.9	45.1	40.8	6.64	50.4	44.0	59.3
Memory bank capacity										
Small memory bank	50	300	9.52	31.3	19.0	17.2	11.83	13.2	11.6	36.2
Default	100	300	**5.19**	**61.2**	**55.0**	**50.5**	**5.59**	**55.4**	**48.4**	**64.6**
Large memory bank	200	300	6.24	58.7	50.2	45.7	6.15	54.3	48.0	61.5

B Additional Results

B.1 Ablations on Memory Token and Bank Capacity

Settings. We evaluate two capacity-related hyperparameters of the episodic memory. The memory token count N_m controls the number of output tokens retrieved by the memory perceiver and injected into the VLM. The memory bank capacity N_b controls the maximum number of frame entries retained after consolidation: a new observation is first appended, and frame merging is invoked only if the bank size exceeds N_b. The default MEMVLN configuration uses $N_m = 300$ and $N_b = 100$. We vary one hyperparameter at a time while keeping the remaining configuration unchanged, including the maximum number of patch tokens per frame $N_f = 100$, and report results on the R2R and RxR Val-Unseen splits following the evaluation protocol in the main paper.

Results. As shown in Tab. B.4, the default setting achieves the best overall performance among the evaluated configurations. Increasing N_m from 300 to 600 degrades all metrics on both benchmarks, showing that a larger memory-token interface does not necessarily translate to stronger navigation capability. One possible explanation is that additional memory tokens introduce redundant or noisy

Table B.5: **Trajectory length distribution after action merging.** We report the percentage of trajectories whose length exceeds a candidate memory bank capacity N_b.

Dataset	> 50 Frames	> 100 Frames	> 200 Frames
R2R	4.7%	≈ 0%	≈ 0%
RxR	45.0%	5.2%	0.3%

Table B.6: **Estimated consolidation pressure with a 50-frame memory bank.** The merge rate is the number of estimated forced merges divided by the number of navigation samples after action merging. Simulation sample counts and the resulting rates are rounded.

Dataset	Nav. Samples	Est. Forced Merges	Merge Rate
R2R	~334K	4,380	~1.3%
RxR	~1.02M	234,453	~23.0%
Joint	~1.35M	238,833	~17.6%

retrieved features, making it harder for the downstream VLM to identify task-relevant historical evidence. Similarly, increasing N_b from 100 to 200 does not yield further improvements. In contrast, reducing N_b from 100 to 50 causes a substantial performance drop, particularly on RxR. Within the evaluated settings, neither the memory-token budget nor the memory bank capacity is a monotonic scaling factor. The only principle is that the bank size should be a properly large number (*e.g.*, 100) that can preserve long-horizon evidence without redundancy.

B.2 Analysis of the Small Memory Bank

Motivation. The main paper notes that similarity-based memory consolidation may inadvertently discard task-relevant observations in excessively long-horizon episodes. The 50-frame memory bank provides a stress test for this limitation. We hypothesize that an undersized bank invokes frame merging too frequently on long trajectories, progressively compressing distinctive visual evidence into an overly coarse episodic summary.

Trajectory Length Distribution. To quantify how frequently frame merging is invoked, we analyze the trajectory length after action merging. As shown in Tab. B.5, only 4.7% of R2R trajectories exceed 50 frames, and almost none exceed 100 frames. In contrast, 45.0% of RxR trajectories exceed 50 frames, while only 5.2% exceed 100 frames. Therefore, setting $N_b = 50$ turns frame merging from an occasional compression mechanism into a routine operation for nearly half of the RxR trajectories.

Consolidation Pressure under Joint Training. The trajectory-level percentages above do not directly reveal how frequently consolidation is invoked across the training samples. To quantify this pressure, for each episode i with length L_i after action merging, we compute the number of excess entries $e_i = \max(L_i - N_b, 0)$. Since each excess entry must be removed to restore the capacity constraint, e_i serves as a proxy for the number of forced frame merges. We then report the merge rate, defined as the total number of estimated forced merges divided by the total number of navigation samples after action merging. As shown in Tab. B.6, with $N_b = 50$, approximately 23.0% of the RxR navigation samples trigger frame consolidation, compared with only 1.3% on R2R. Under joint training, RxR contributes 98.2% of all estimated forced merges, yielding an overall merge rate of approximately 17.6%. Therefore, the shared memory module is frequently exposed to repeatedly consolidated RxR states during joint training.

R2R-only Diagnostic. We further conduct an R2R-only diagnostic experiment with $N_b = 50$. As shown in Tab. B.7, the 50-frame bank achieves 45.5% SR when trained and evaluated only on R2R, substantially recovering from the 19.0% SR obtained under joint R2R/RxR training. This diagnostic does not show that a 50-frame bank is optimal for R2R, but it shows that the small-bank configuration does not exhibit the same severe failure mode when rarely exceed the bank capacity.

Table B.7: **R2R diagnostic results with a 50-frame memory bank.** The R2R-only run is substantially more stable than joint R2R/RxR training with the same bank capacity.

Training Setting	NE↓	OSR↑	SR↑	SPL↑
Joint R2R/RxR	9.52	31.3	19.0	17.2
R2R-only diagnostic	6.40	53.3	45.5	41.7

Table B.8: **Real-world evaluation instructions.** We use the instruction IDs in subsequent results.

ID	Instruction
I1	Now you are facing a brown curtain. Turn right and walk pass an armchair towards the fire extinguisher. Turn left and walk toward the glass door in front of you to the right, that's your destination.
I2	Now you are facing the reception. On your left you should see some brown sofas. Walk past the sofas to the handwriting on the wall. Stop at the door on your right.
I3	Now you are in a corridor. Keep moving forward. Once you pass the bookshelf on the right side, turn right and go to the sink in the corner.
I4	Go straight towards the glass windows and past the sofas on your right. After you reach the windows, turn right. Keep moving forward and stop at the second door on your right.
I5	Now you are in a corridor. On your left, there are some white cabinets. Go past them and then turn left. You should see a red fire extinguisher on the ground. That box is your destination.
I6	Now you are in a lab. On your right there are some wooden stakes. Move forward and pass the chairs in front of you. You should see a robot sitting on a chair. Walk past the robot and stop at the bucket next to the lamp.

Discussion. The trajectory statistics, consolidation-pressure estimate, and R2R-only diagnostic are consistent with our hypothesis that aggressive repeated frame merging is a likely failure mode of an undersized memory bank. In particular, the pressure imbalance suggests that, during joint training, the shared memory module is frequently exposed to heavily consolidated RxR states, which may make the learned memory representations less reliable on both benchmarks. This diagnostic shows that the severe R2R degradation with $N_b = 50$ is not caused by the 50-frame bank alone on R2R-like trajectories. Instead, when combined with the merge-rate analysis, it indicates that the failure is likely driven by the high consolidation pressure introduced by RxR during joint training. Nevertheless, this analysis provides evidence for why the default $N_b = 100$ setting is effective: it preserves sufficient capacity for the large majority of trajectories while still bounding the cost of the memory perceiver. Adaptive consolidation strategies for extremely long episodes remains an important future direction.

B.3 Real-world Results

Evaluation Protocol. We provide a more detailed breakdown of the real-world G1 evaluation summarized in Figure 3 of the main paper. We evaluate four variants: Base, the VLM baseline; +Mem, the baseline augmented with the memory module; MemVLN, the full model; and MemVLN-PT, the full model after real-world post-training with 24 demonstrations. The evaluation uses six held-out real-world instructions, listed in Tab. B.8. Each method is evaluated with five independent runs per instruction, resulting in 30 runs per method. In addition to the success rate (SR), we report the collision rate (CR). CR measures the fraction of runs that involve physical contact, contact-induced blockage, or contact-induced termination during execution; it is independent from SR and reflects contact safety rather than task completion.

Deployment Setup. During real-world deployment, policy inference runs on a host workstation equipped with a single NVIDIA RTX 4090 GPU. RGB observations captured by the robot-mounted Intel RealSense D455 are transmitted to the host over Wi-Fi; after the host completes model inference, the predicted action is sent back to the G1 robot for execution.

Overall Performance. The detailed real-world results highlight both the benefit and the current limitation of post-training. As shown in Tab. B.9, Base succeeds in 5 of 30 runs, corresponding to 16.7% SR. Adding the memory module increases success to 9 of 30 runs (30.0% SR), and the full

Table B.9: **Overall real-world results.** Each method is evaluated with five independent runs per instruction, resulting in 30 runs per method. CR denotes the collision rate.

Method	Runs	Successes	SR↑	CR↓
Base	30	5	16.7	56.7
+Mem	30	9	30.0	66.7
MemVLN	30	12	40.0	**46.7**
MemVLN-PT	30	**14**	**46.7**	**46.7**

Table B.10: **Per-instruction real-world results.** Each cell reports SR / CR (%) over five independent runs for the corresponding instruction and method. Bold marks the best nonzero SR and lowest CR in each row.

ID	Base	+Mem	MemVLN	MemVLN-PT
I1	20 / 40	40 / 60	**80** / **20**	60 / 80
I2	0 / 60	0 / 60	**20** / 60	**20** / **40**
I3	40 / 60	**80** / 60	40 / **40**	**80** / 60
I4	0 / 100	0 / 100	0 / **60**	0 / **60**
I5	20 / **20**	0 / 40	20 / 40	**60** / 40
I6	20 / 60	60 / 80	**80** / 60	60 / **0**

MemVLN model further improves success to 12 of 30 runs (40.0% SR). After real-world post-training with 24 demonstrations, MemVLN-PT succeeds in 14 of 30 runs (46.7% SR), giving two additional successful runs over MemVLN. Thus, real-world demonstrations provide a measurable adaptation signal, but the improvement is modest in this held-out evaluation. At the same time, CR remains unchanged at 46.7% for MemVLN and MemVLN-PT, indicating that the current post-training setup improves task completion more than contact safety. This motivates further exploration of real-world adaptation strategies with more data, different objectives, or stronger safety-aware supervision.

Instruction-level Analysis. The per-instruction results in Tab. B.10 explain why the post-training gain remains modest. Relative to MemVLN, MemVLN-PT improves SR on I3 and I5, remains unchanged on I2 and I4, and decreases on I1 and I6. CR also does not consistently decrease across instructions: post-training lowers CR on I2 and I6, but increases it on I1 and I3. The clearest remaining weakness is I4, where all four methods obtain 0% SR. This instruction requires selecting the second door on the right after passing an intermediate door, indicating that ordinal landmark grounding remains brittle in real-world deployment.

Remaining Challenges. The detailed results also clarify where real-world adaptation still has room to improve. First, collision remains a major challenge: even the best-performing variants, MemVLN and MemVLN-PT, have a CR of 46.7%. Second, the 0% SR on I4 across all methods shows that some language-grounding patterns remain unsolved, especially instructions that require distinguishing an ordinal landmark such as the "second door on your right." This failure suggests a broader challenge for real-world VLN: complex instructions, fine-grained landmarks, and routes with multiple similar candidate goals may require more targeted adaptation than the current behavior cloning setup with 24 demonstrations. Future work should therefore explore stronger contact-aware control and richer supervision for ordinal and fine-grained spatial grounding.

B.4 More Qualitative Results

Simulation Results Figure B.1 and B.2 provide representative qualitative rollouts on R2R-CE and RxR-CE, respectively. These examples qualitatively complement the aggregate metrics by illustrating cases in which MEMVLN follows multi-step instructions over extended trajectories where episodic context can be useful.



Figure B.1: **Qualitative results of MEMVLN on R2R-CE.**

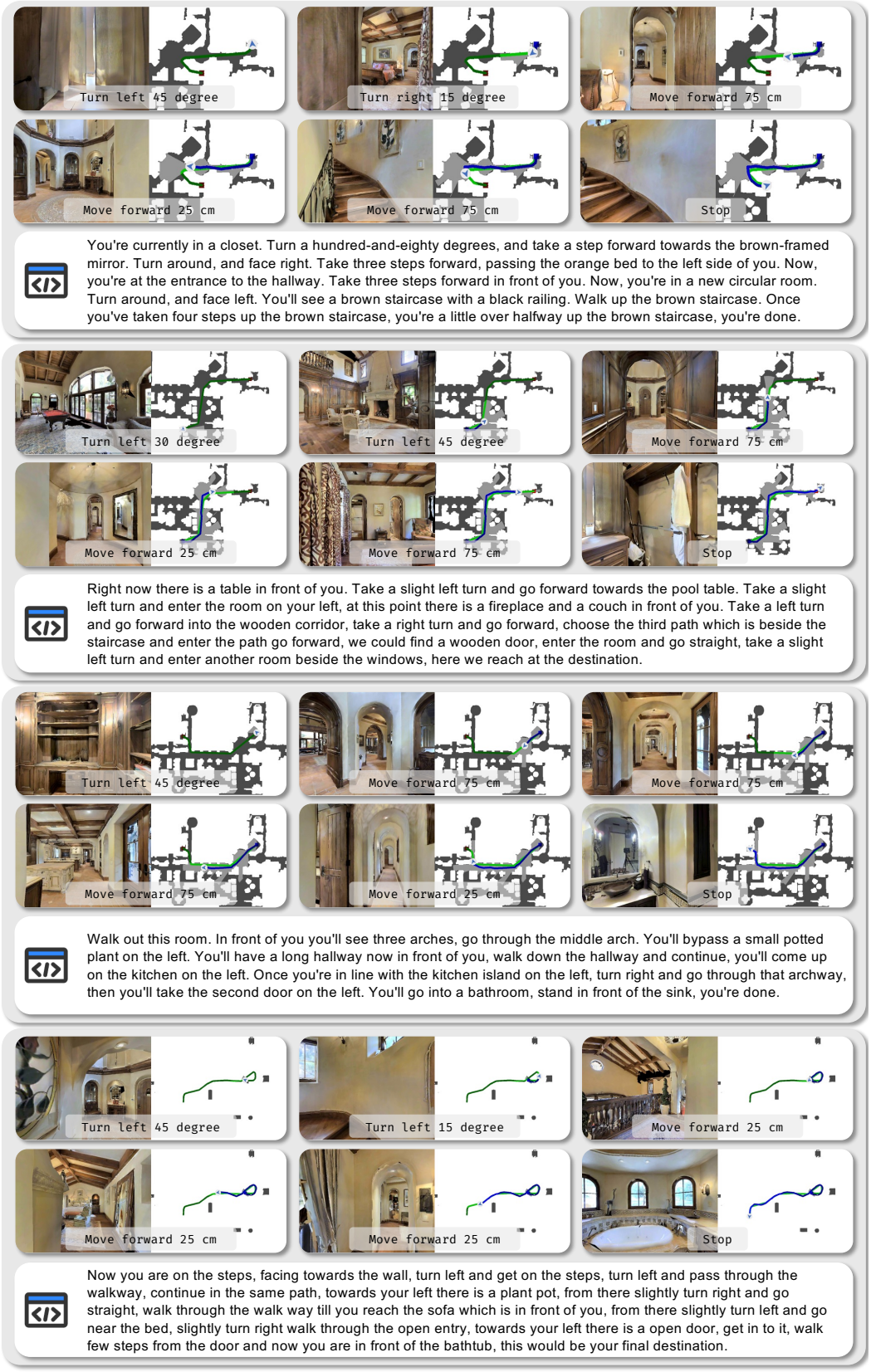


Figure B.2: **Qualitative results of MEMVLN on RxR-CE.**

References

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